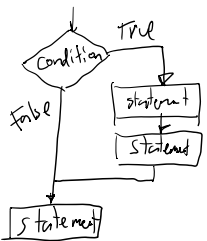


Decision Structures

If statement

↳ if condition is True then run the if-block,



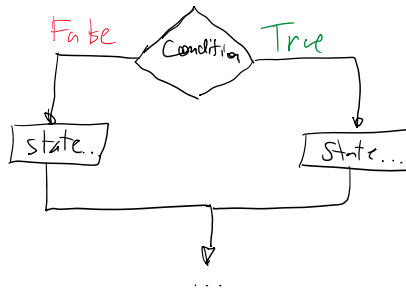
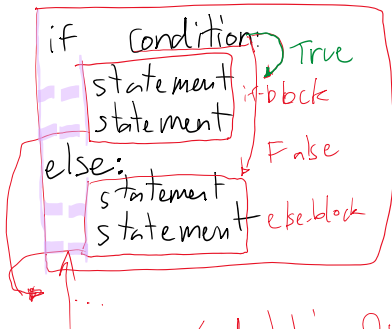
if condition: ^{True/False}
 — statement ^{True}
 — statement ^{if-block = True/False}
 statement ^{← True/False}

Ex1
 cond = (7 == 6) ^{False}
 if cond: ^{False}
 print('A')
 print('B')
 print('Done') → Done

Ex2
 if 10 != 5: ^{True}
 print('A') → A
 print('B') → B
 if 10 < 3: ^{False}
 print('C')
 print('Done') → Done

If-else Statement

↳ if condition is True then run the if block, if False then run the else block



space / indentation is used to identify the if / else-block

Ex1
 print('A') → A
 if 5 > 3: ^{True}
 print('B') → B
 else:
 print('C')
 print('Done') → D

print('A') → A
 if 3 < 1: ^{False}
 print('B')
 else:
 print('C') → C
 print('Done') → Done

Ex Donut shop

- If user donut on user
- donut from 20 min to 10 min → 11:00 am to 12:00 pm

* and bina 1 620
 * or bina 1 620 620 620

ex1 $X = (\text{True and True}) \rightarrow \text{True}$
 $Y = (\text{True and False}) \rightarrow \text{False}$
 $\text{print}(X, Y) \rightarrow \text{True False}$

ex2 $\text{cond1} = ((3 \neq 2) \text{ or } (4 \neq 5)) \text{ and } (7 > 2)$
 $\text{cond2} = (\text{True or False}) \text{ or } (\text{True and False})$
 $\text{if } \text{cond1} \text{ and } \text{cond2}:$
 print('A')
 else:
 print('B')
 print('Done')

$\text{salary} = \dots$
 $\text{year} = \dots$
 $\text{if } (\text{salary} \geq 20000) \text{ and } (\text{year} \geq 2):$
 print("Approve")
 else:
 $\text{print("Not approve")}$

Check interval

person given an age score x or y 1200?
 like with teenager an age score 13 to 18
 $\text{age} \geq 13 \text{ and } \text{age} \leq 18 \rightarrow 13 \leq \text{age} \leq 18$

$\text{age} = \text{int}(\text{input("age:")})$
 $\text{if } \text{age} \geq 13 \text{ and } \text{age} \leq 18:$
 print("Teenager")

$\text{age} < 13 \text{ or } \text{age} > 18$

Grading Question

Give the score:

$\text{if } \text{score} \geq 80 \rightarrow 'A'$
 $\text{if } 80 > \text{score} \geq 70 \rightarrow 'B'$
 $\text{if } 70 > \text{score} \geq 60 \rightarrow 'C'$
 $\text{if } \text{score} > 60 \rightarrow 'D'$

• $\hat{r}_{uv}$ score on user u .

score = float(input()) 90

```
# If nested if else
score = float(input("..."))
```

```
if score >= 80:
    print('A')
elif score >= 70:
    print('B')
elif score >= 60:
    print('C')
else:
    print('D')
```

```
if score >= 80: 80 >= 80 ✓
    print('A') → A
if score >= 70 80 >= 70 ✓
    print('B') → B
if score >= 60 80 >= 60 ✓
    print('C') → C
else:
    print('D')
```

* $a \leq b$ คือ $b \geq a$ ใน $\mathbb{N}$ และ $b \geq a$ ใน $\mathbb{Z}$ หรือ $b \geq a$ ใน $\mathbb{R}$
 * $a \leq b$ คือ $b \geq a$ ใน $\mathbb{N}$ และ $b \geq a$ ใน $\mathbb{Z}$ หรือ $b \geq a$ ใน $\mathbb{R}$
 * $a \leq b$ คือ $b \geq a$ ใน $\mathbb{N}$ และ $b \geq a$ ใน $\mathbb{Z}$ หรือ $b \geq a$ ใน $\mathbb{R}$

* No H-Cl-E-Be
but a structure
↳ $\text{O}=\text{C}=\text{O}$ bond
by $\text{O}=\text{N}-\text{O}$ bond
st. v. n. g.!

```
X = 5
if x > 10:
    print('A')
elif x > 7:
    print('B')
elif x > 3:
    print('C')
else:
    print('D')
```

The image shows the execution flow for the above code. Red arrows indicate the evaluation of conditions: $5 > 10$ is False, $5 > 7$ is False, and $5 > 3$ is True. A blue arrow points from the true condition to the 'C' branch. Another blue arrow points from the 'else' branch to the label 'Done'.

~~print('Done')~~ → Done

Decision Structures

if condition:

statement

statement

True

if condition:

statement

else:

statement

True

False

if condition 1:

statement

elif condition 2:

statement

elif condition 3:

statement

else:

statement

T

F

T

F

T

F

Comparison Operators: $>$, $<$, $>=$, $<=$, $==$, $!=$

Boolean Operators: and, or, not